3D Slicer Bounding Box Navigator

Quick User Guide

First-Time Setup

1. **Extract the ZIP**: Right-click the downloaded zip file → **Extract All...**
2. Open **3D Slicer**.
3. In the menu, go to **Edit** → **Application Settings** → **Modules**.
4. Under **Additional module paths**, click **Add** and select the **`BoundingBoxNavigator`** folder. *Tip:* if the settings window is small, the **Add** button is hidden behind a **>>** button to the right of the **Additional module paths** box; click **>>** to reveal it.
5. Click **OK** and restart Slicer.

macOS: Drag Slicer from the disk image into **Applications** (do not run it from the disk image). The first time, right-click Slicer → **Open** → **Open**. Move this tool's folder to your home folder (Finder: Cmd+Shift+H), not Downloads, Desktop, or Documents, or the module disappears after a restart. Then double-click `launch_mac.command`.

How to Annotate

1. **Open the Module**
   - In the top toolbar, go to: **Modules** → **Annotation** → **Bounding Box Navigator** to display the panel on the right.
2. **Select Folders & Scan**
   - Set **Input Scans Folder** (where your CT scans are) and **Output Annotations Folder**.
   - Click **Scan Folder**. The first unreviewed case loads automatically.
   - Each slice view shows the **original** scan for that plane when the file exists; otherwise it reconstructs from the axial volume (a warning is shown).
3. **Draw Bounding Boxes**
   - Choose a **Finding type** (default: PM nodule). Then press **B** (or click **+ Add Bounding Box**).
   - **Click and drag** across the lesion on any slice view (axial, coronal, or sagittal).
   - Drag the colored handles on the box to resize it in width, height, and depth.
   - Scroll through slices with your mouse wheel to check coverage.
   - Use the large **slice thickness** buttons (3 mm, TS, . . .). The box stays in patient coordinates, so the same box is used on TS, 2 mm, 3 mm, or 5 mm. A pop-up warns if those series do not share one coordinate system. For TS (or any thickness without native COR/SAG), coronal and sagittal show reconstructions of that axial scan.
   - Press **B** again for each additional lesion.
   - *To hide or show a box*: untick / tick the **Show** checkbox in front of it in the box list. Hidden boxes are still saved; this only hides them while you look at the scan.
   - *To remove a box*: click it in the list and press **Delete Selected**.
4. **Notes / PCI Score**
   - Type the PCI score or clinical comments in the **Notes / PCI** field (optional).
   - Optional: in **4. Segmentation**, load a `.seg.nrrd` or `.nii.gz` overlay (`.nii.gz` is imported as a labelmap, not a CT volume).
5. **Next Scan**
   - A box is written when you release the mouse after drawing or resizing it. The scan stays in the queue until you click **Next**.
   - Click **Next** (large green button) to mark the case complete and load the next unreviewed scan.
   - Each box is saved as `R_1.mrk.json`, `A_1.mrk.json`, . . . inside a folder named after the case ID.
   - *If there are 0 lesions (PCI 0)*: click **Next** and confirm with **Yes**.

Mouse & Keyboard Shortcuts

Shortcut / Control	Action
B	Draw a new bounding box
Mouse Wheel	Scroll through CT slices
Middle Click + Drag	Pan view
Right Click + Drag	Zoom in / out
Click row in box table	Centers all slice views on that lesion
Show checkbox (box table)	Hide / show that bounding box in the views; hidden boxes are still saved
Skip completed (checkbox)	Checked by default. Uncheck to browse or edit past cases
Next	Save this case and open the next one
Slice thickness buttons	Switch 3 mm / TS / other; the box stays on the same tissue; COR/SAG reconstruct from axial if ne
Finding type	PM nodule (R) default; also ascites, omental cake, stranding, lymph node

Troubleshooting: folder layout

The case name comes from the **filename**, not from the folder the file sits in. Subfolders are searched. Two patterns are accepted, and both may sit in the same folder:

```
{case_id}_{AX|COR|SAG}_{phase}_{thickness}.nii.gz
{case_id}_{scan number}_{AX|COR|SAG}_{phase}_{thickness}.nii.gz
```

- **case_id**: one token, no underscores (example: `1001`).
- **scan number**: optional digits (`2` or `02`). When it is absent, the scan is number 1, so `1001_AX_...` and `1001_1_AX_...` are one case, `1001`. Scan 2 and higher are their own case, such as `1001_2`.
- **Plane** must be `AX`, `COR`, or `SAG`. Every case needs an `AX` file. A missing `COR` or `SAG` is reconstructed from the axial scan, and a warning is shown.
- **Phase** (for example `pvp`) is read and then ignored for grouping. **Thickness** is `3mm`, `TS`, `2mm`, and so on.

Layouts that work

```
scans/                                  annotations/   (created on Save)
  1001_AX_pvp_3mm.nii.gz                  1001/
  1001_COR_pvp_3mm.nii.gz                   R_1.mrk.json
  1001_SAG_pvp_3mm.nii.gz                   _annotation_done.json
  1001_2_AX_pvp_3mm.nii.gz                1001_2/
  1001_2_COR_pvp_3mm.nii.gz                 R_1.mrk.json
  patient_1004/                             _annotation_done.json
    1004_AX_pvp_3mm.nii.gz
```

`1001_*` is case `1001`, scan 1. `1001_2_*` is case `1001_2`. `patient_1004/` does not matter: `1004_AX_...` is case `1004`. On Save, boxes go to a folder of that same name. An existing folder is reused when it is named exactly `1001` or `1001_2`, or when the name contains that id (`Scan 1001`). The folder `1001_2` is not used for case `1001`.

When the scan does not line up

What you see	How to align the files
Two scans of one patient become one case, or boxes are saved in the wrong folder	Put the scan number in the filename: `1001_2_AX_pvp_3mm.nii.gz`. A folder named `1001_2` is not enough.
A warning names a contrast phase conflict, or a file is ignored as a duplicate	Keep one scan per case id and scan number. Give another scan its own number, such as `1001_2_AX_art_3mm.nii.gz`.
The case is called `image` or `volume`, not `1001`	The id is only on the folder. Rename the file so the id is in the name. `scans/1001/image.nii.gz` is case `image`.
A warning says folder `1001` contains a file for case `1002`	Move that file to its own folder. One case folder should not hold another case.
A case never appears	Add an axial file (`AX`). Coronal and sagittal files alone are not a case.
Coronal and sagittal are reconstructions, and the case name is the whole filename	The name does not match the pattern above. Keep the order: case id, optional scan number, plane, phase, thickness.
Scan Folder finds nothing	Point **Input Scans Folder** at the folder of `.nii.gz`, `.nii`, `.nrrd`, `.nhdr`, `.mha`, or `.mhd` files, not at the annotations folder.

A file that matches neither pattern is still loaded as its own case. A markup file that cannot be read is kept and listed. A box on disk does not complete the case; Next writes the completion marker. Previous saves a changed case before leaving. Reload can drop edits that are not written yet. A geometry pop-up lists orientation, in-plane spacing, in-plane origin, or a box outside a reconstructed view.